

The Transformer Revolution

Part 2

LIN 313 Language and Computers
Fall 2025

Objectives

- Review attention
- Contextual word embeddings
- Maybe: training a chatbot with RLHF

What do we need to understand ChatGPT?

- language modeling task
- probability
- perceptrons
- neural networks
- deep learning (Multi-layer perceptrons)
- attention
- contextual embeddings
- pre-training
- decoder-only model (GPTs, all LLMs)
- context-window
- RLHF (reinforcement learning with human feedback)

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Translation, again

How do you translate 'leg'
into French?

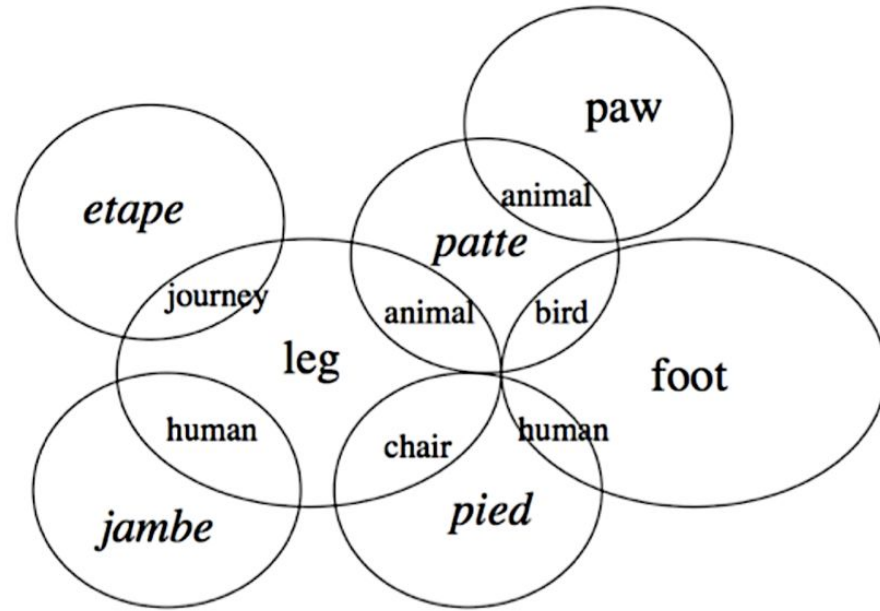
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- Depending on the context, English '**leg**' could be translated into French '**jambe**' (human), '**pied**' (chair), '**patte**' (animal), or '**étape**' (journey).

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27

source: Jason Baldridge

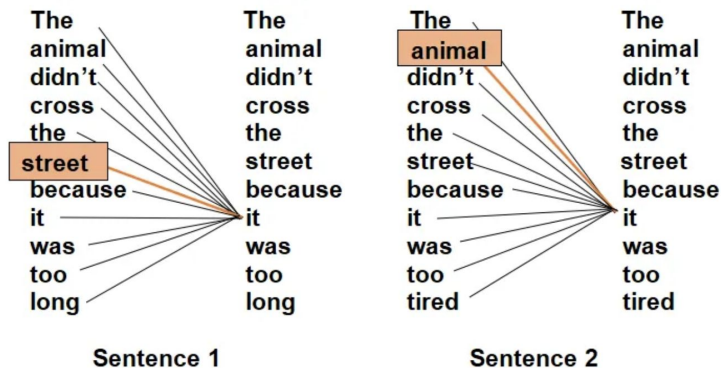
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What is Attention?

Intuition: At different steps, let a model "focus" on different parts of the input.

An attention mechanism is a part of a neural network. At each decoder step, it decides which source parts are more important.

Q: How does this help with the overlapping meanings problem?



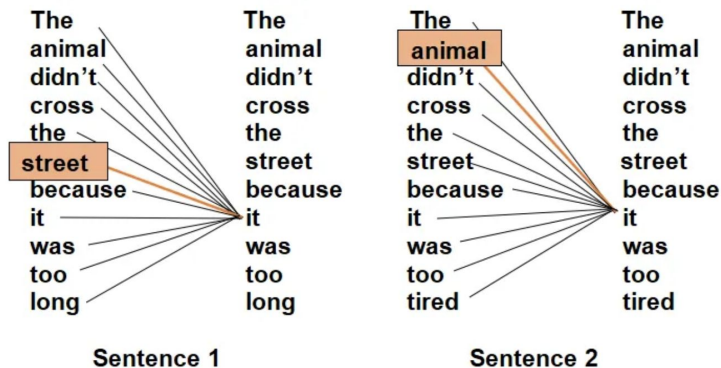
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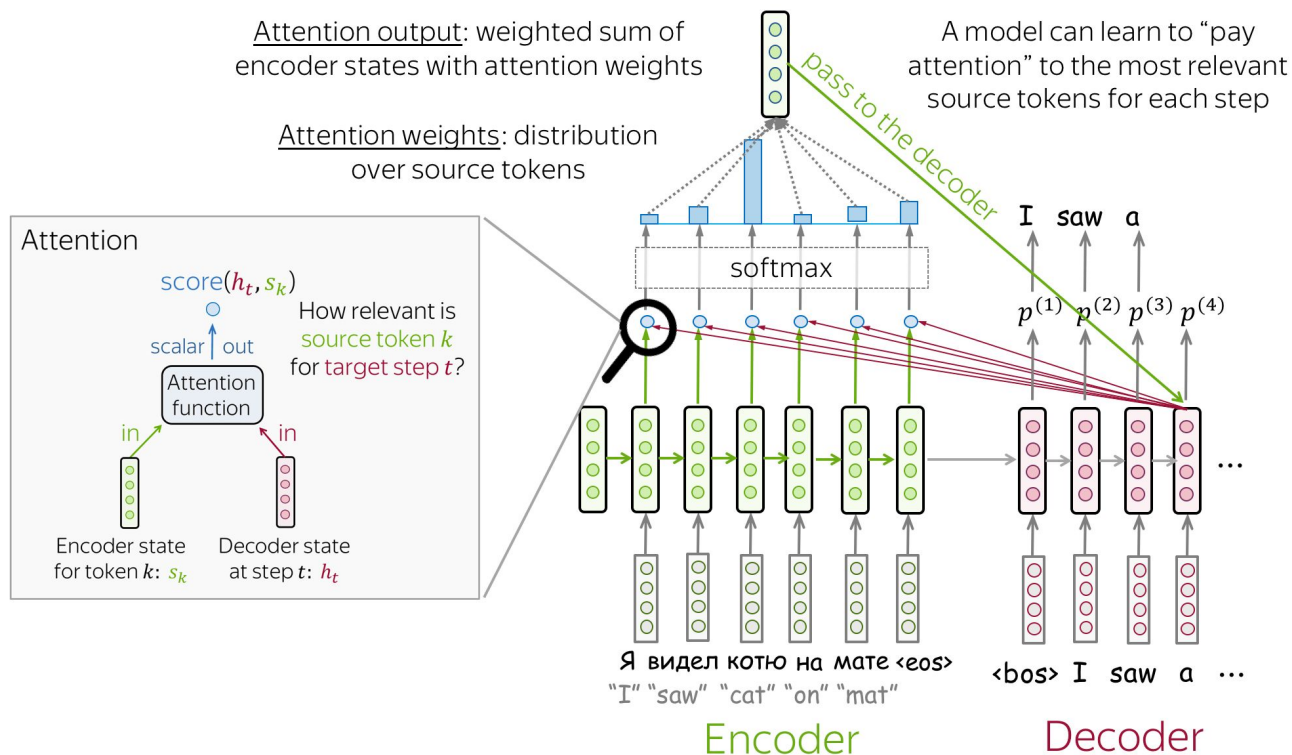
A: The model can attend to the other words in the input that disambiguate which translation we are looking for.



Information flow in the Transformer with Attention



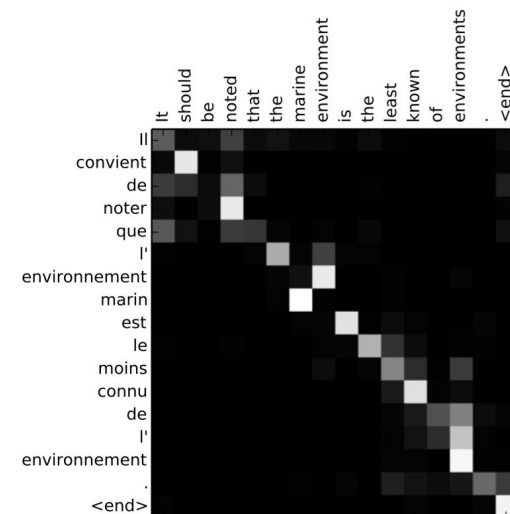
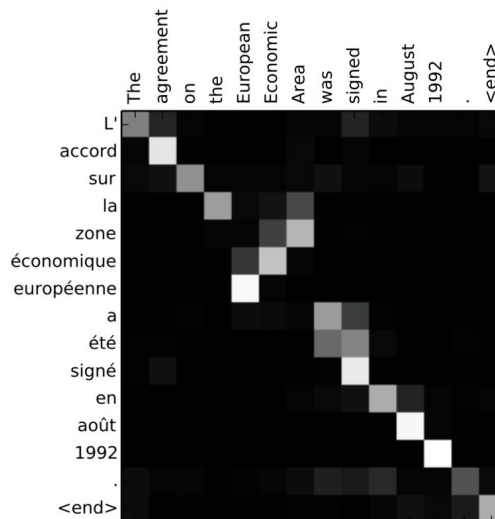
What does the attention mechanism "look like"?



To see the weights change step-by-step https://lena-voita.github.io/nlp_course/seq2seq_and_attention.html

What does Attention learn?

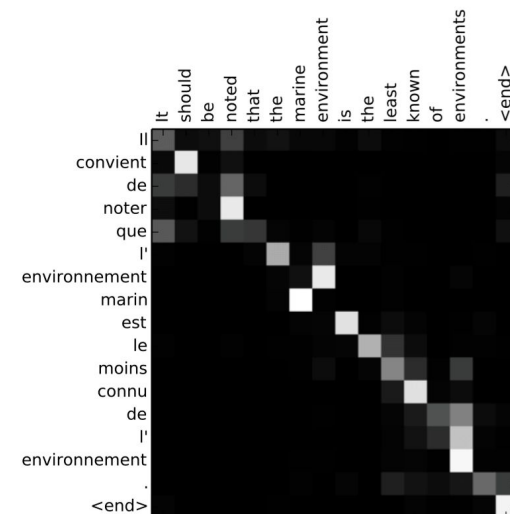
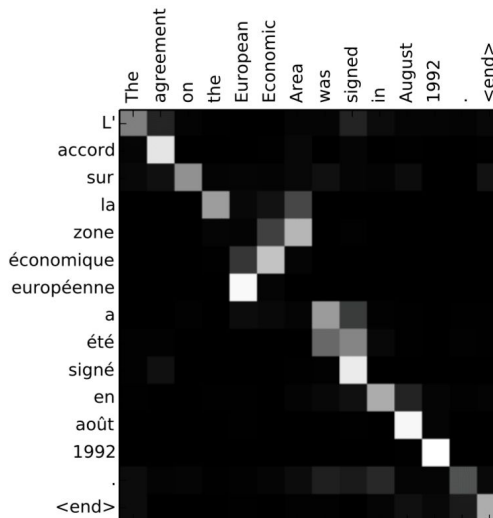
Here are **attention maps** for translating from English to French. They visualize the attention weight on the input tokens at each stage in the decoding. From the examples, we see that attention learns (soft) alignment between source and target words.



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How do adjectives work in French?

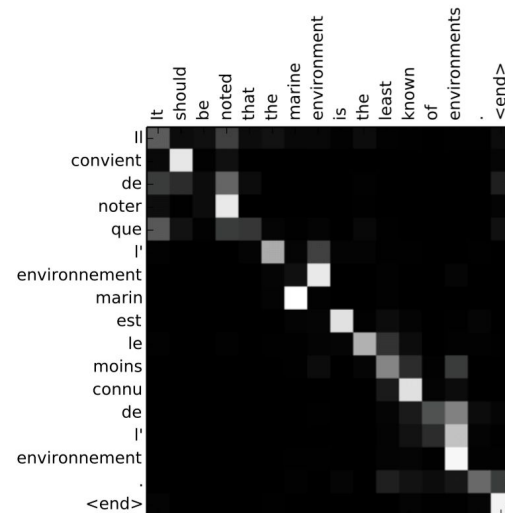
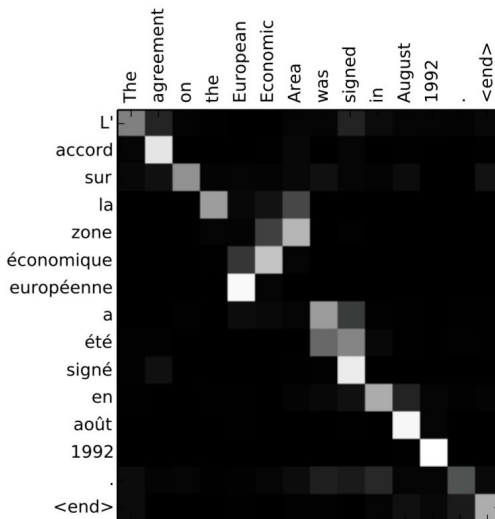


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How do adjectives work in French?

Where is there a one-to-many relationship from English to French?



We also need to pay attention in one language

One word can have many meanings

"Channel 4 news: always first on the scene for the **scoop**."

"Let's go down to the ice-cream shop for a **scoop**."

"So what's the **scoop** on the new kid?"

- Homonymy: same word, totally unrelated meanings
- Polysemy: multiple related meanings

Types of Polysemy

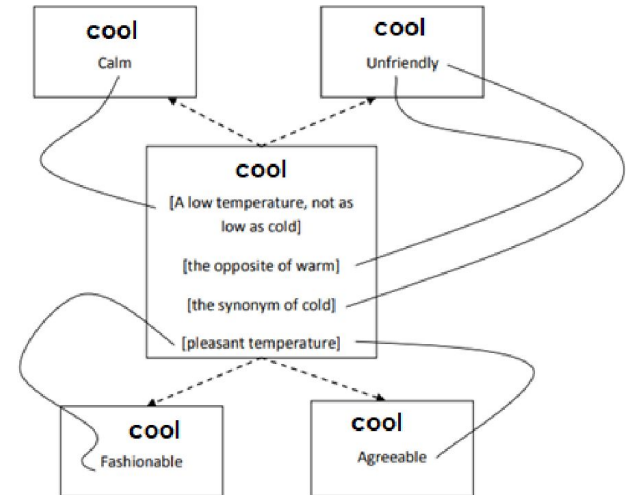
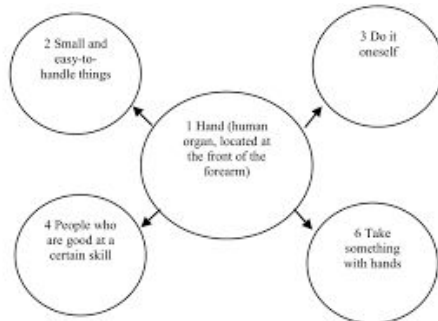
1. Linear/Chain Polysemy: Meanings connected in a sequence through metaphorical extension

- "head": body part → leader → top of page → foam on beer
- Each meaning evolved from the previous one

Types of Polysemy

2. Radial Polysemy: Multiple meanings radiating from a central, prototypical sense

- "hand": body part (central) → help ("give me a hand") → worker ("all hands on deck") → pointer on clock → handwriting
- All meanings relate back to the core concept



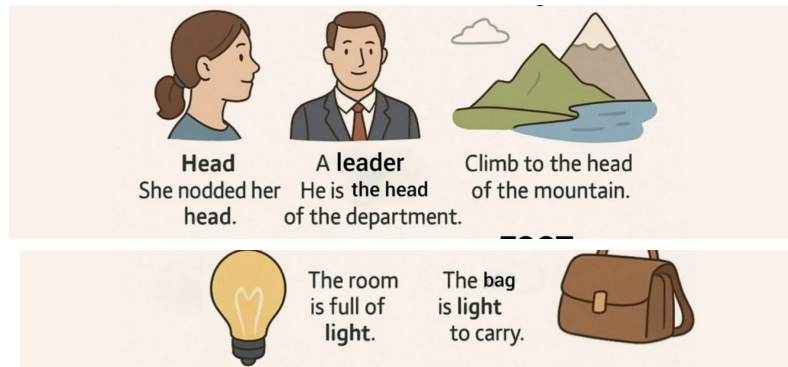
Types of Polysemy

3. Metaphorical Polysemy: Abstract meanings derived from concrete ones

- "see": visual perception → comprehend ("I see what you mean")
- KNOWLEDGE is SIGHT
 - this one is baked into English very deep
 - **conceptual metaphors** organize whole domains of knowledge in terms of other domains (Lakoff & Johnson 1980)

wise (adj.)

"having the power of judging or discerning rightly," Old English *wis* "learned, sagacious, cunning; sane; prudent, discreet; experienced, practically knowing;" from Proto-Germanic **wissaz*, reconstructed to be from past-participle adjective **wittos* of PIE root ***weid-** "to see" (hence "to know;" compare **wise** (n.)).



4. Metonymic Polysemy: Meanings through contiguity/association

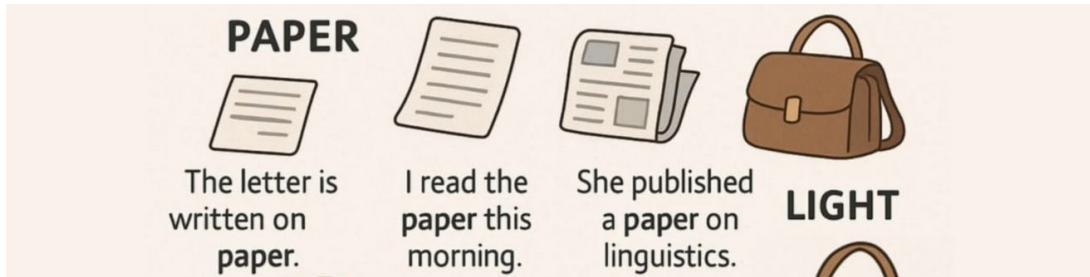
- "crown": physical object → monarchy → tooth cap
- "dish": container → food in it ("tasty dish")



Types of Polysemy

5. Logical Polysemy: Systematic alternations across word classes (James Pustejovsky 1990)

- "book": physical object / informational content
- "chicken": animal / meat
- "newspaper": organization / physical paper



Types of Polysemy

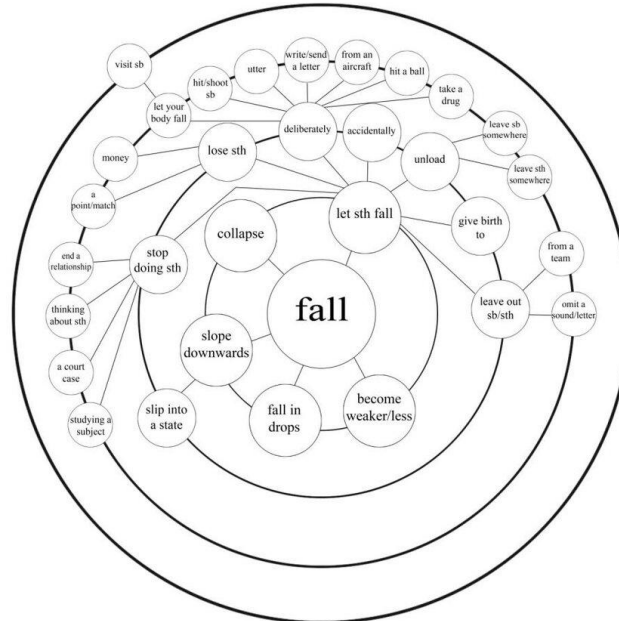
In practice, these are not distinct types. We can see many types of polysemy at work in a single word.

regular:

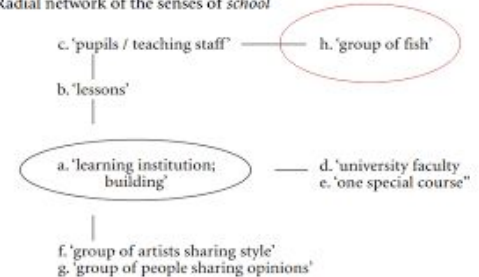
- *I love fall* (the season)
- *See you in the fall* (time period)

metaphorical

- *fall from grace*
- *fall from a tree*



Radial network of the senses of *school*



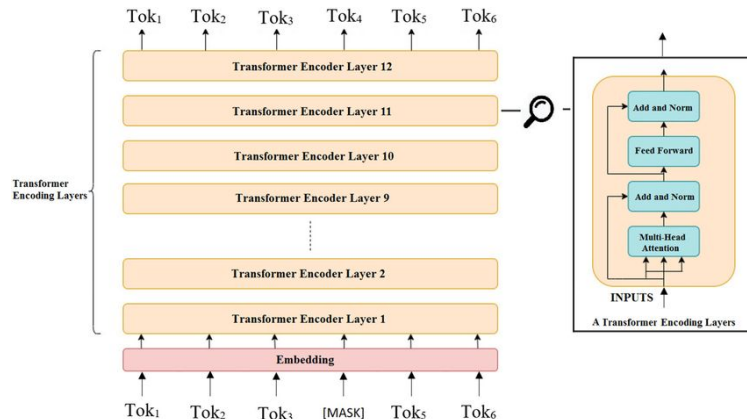
The Next innovation: BERT

Intuition: Do we get better word vector representations if we use attention on just one language?

BERT (Devlin et al. 2017) (Google)

- Task: Masked Language Modeling (MLM)
 - given a sentence, predict the missing word
- architecture
 - **encoder only:** Just use the first half of the encoder-decoder.
 - attention over the input sequence (self attention)
 - pros:
 - gives us a good representation of the entire that can be used for classification tasks.
 - good representations of words
 - cons:
 - not good for generation (BERT looks both ways so we have to know what's coming next).
- 340 million parameters.
- 24 layers

https://cogcomp.seas.upenn.edu/page/demo_view/BERT



roberta-base ▾

Example Inputs ▾

The sky at sunset looked like a @ flower.

[MASK]

('beautiful', 10.219)

('red', 9.042)

('pink', 8.95)

('yellow', 8.887)

('white', 8.872)

('green', 8.571)

Transformer Revolution

BERT was quickly followed by many other **pretrained language models** with small variations. ERNIE, ELMO... "sesame street" dominated the scene. The variations were minor.

Research in two stages:

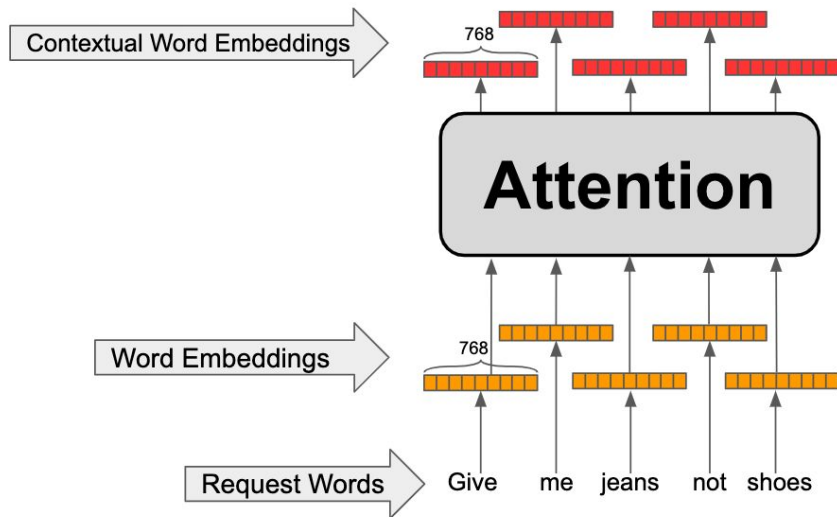
- Pretrain (**un/semi-supervised**) - train a large neural network on a general purpose task (like masked word prediction or next word prediction) using a lot of unannotated text
- Finetune (**supervised**) - continue to train the model on a smaller dataset annotated with the data for your specific problem (inference prediction, winograd schema solving, sentiment analysis, etc. etc. (superv



Contextual Word Embeddings

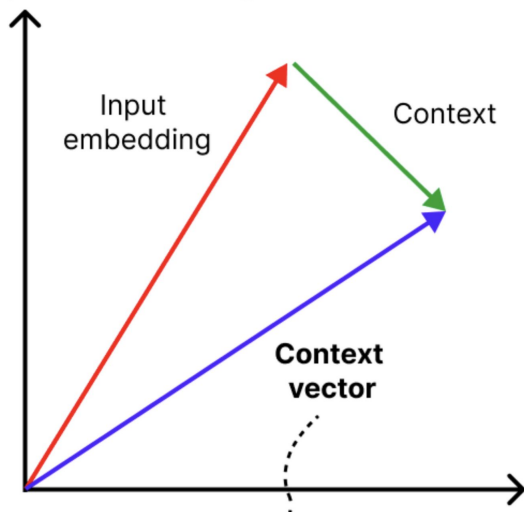
Attention means that as information flows through the network, the embeddings downstream of any one token also incorporate information from other tokens in the input.

- The first layer of embeddings represents *shoes* as a type-level, context-free vectors (like word2vec).
- The attention mechanism **modulates** the 'static' embedding for a *shoes* according to the embeddings for the neighboring words.



Contextual Word Embeddings

Attention mechanism: Convert all input embeddings to context vector

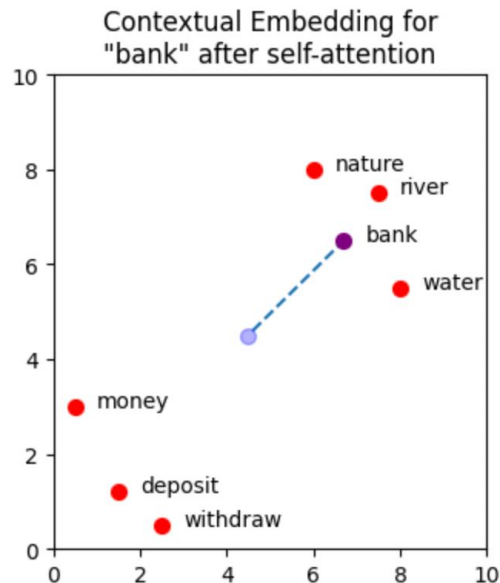
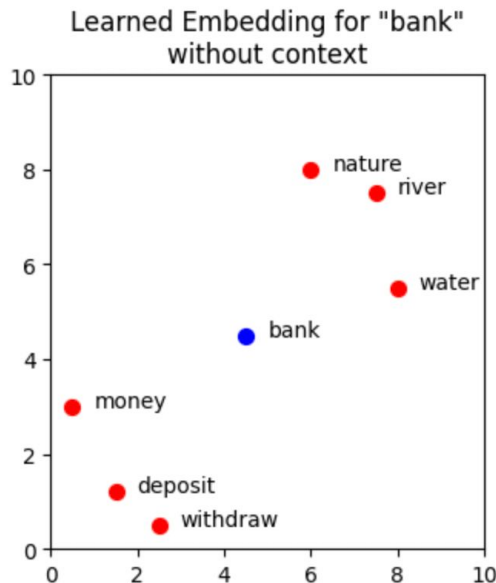


This is much richer. Because it has information about neighboring tokens.

Contextual Word Embeddings

Where we used to have one embedding vector per word *type*, we now have one embedding per word *token*.

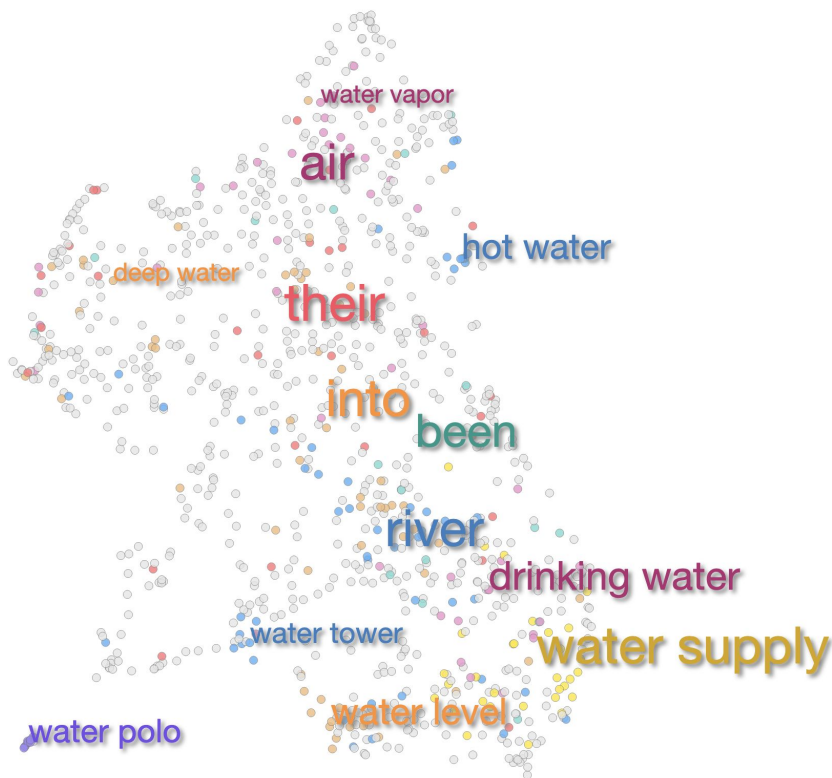
You can think of attention as 'looking up' which meaning of the word we are dealing with, or 'soaking up' information about the context



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demo:

<https://storage.googleapis.com/bert-wsd-vis/demo/index.html?#word=water>